Your grade: 96.66%

Your latest: **96.66%** • Your highest: **96.66%** • To pass you need at least 80%. We keep your highest score.

Next item

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding metrics for model comparison, loss functions, and image preprocessing, but there is room for growth in understanding the training process of neural networks. Before retrying this assignment, we recommend focusing on these key areas:

- **Multi-Class 7: Training Loop**: In Question 2, your understanding of the processes involved in training a neural network model was incomplete. Try reviewing this item to refresh your memory on the steps involved in initializing gradients, performing forward and backward passes, updating weights, and monitoring the progress of losses.

Good job on the rest of the questions!

Retry assignment

1. What metric would you use to compare the performance of a naive classifier with more advanced models in a multi-label classification problem? 1 / 1 point

- (•) Accuracy
- ( ) F1 Score
- ( ) Recall
- ( ) Precision

Correct
Correct! Accuracy is a common metric used to compare the performance of different models.

2. Which of the following processes are involved in training a neural network model? 0.8333333333333334 / 1 point

- [ ] Model Deployment
- [ ] Initializing Gradients
- [x] Forward Pass

Correct
Correct! Forward pass is an essential step in training a neural network model.

- [x] Backward Pass

Correct
Correct! Backward pass is an important step in updating the weights of the model.

- [ ] Data Augmentation
- [x] Updating Weights

Correct
Correct! Updating weights is part of the training process of a neural network model.

You didn't select all the correct answers

3. When implementing a binary image classification using a Convolutional Neural Network (CNN), which loss function is typically used? 1 / 1 point

- ( ) Categorical Cross-Entropy
- ( ) Hinge Loss
- ( ) Mean Squared Error
- (•) Binary Cross-Entropy

Correct
Correct! Binary Cross-Entropy is the standard loss function for binary classification tasks.

4. Which of the following are essential steps in the image preprocessing pipeline for a Convolutional Neural Network (CNN)? 1 / 1 point

- [ ] Image Segmentation
- [x] Cropping

Correct
Correct! Cropping can be used to focus on the relevant parts of an image and remove unnecessary background.

- [ ] Histogram Equalization
- [x] Resizing

Correct
Correct! Resizing is a crucial step to ensure all input images have the same dimensions.

- [x] Normalization

Correct
Correct! Normalization helps scale the pixel values to a standard range, which can speed up the training process.

5. What is the role of Fast Fourier Transformation (FFT) in treating audio files as images for CNNs? 1 / 1 point

- ( ) To compress audio signals for faster processing.
- ( ) To enhance the resolution of audio signals.
- ( ) To filter out background noise from audio signals.
- (•) To transform time-domain audio signals into frequency-domain representations.

Correct
Correct! FFT transforms time-domain audio signals into frequency-domain representations, making it easier to treat them as images for CNNs.